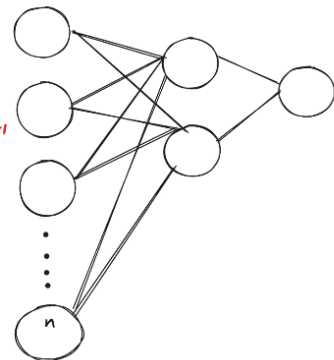


RNN

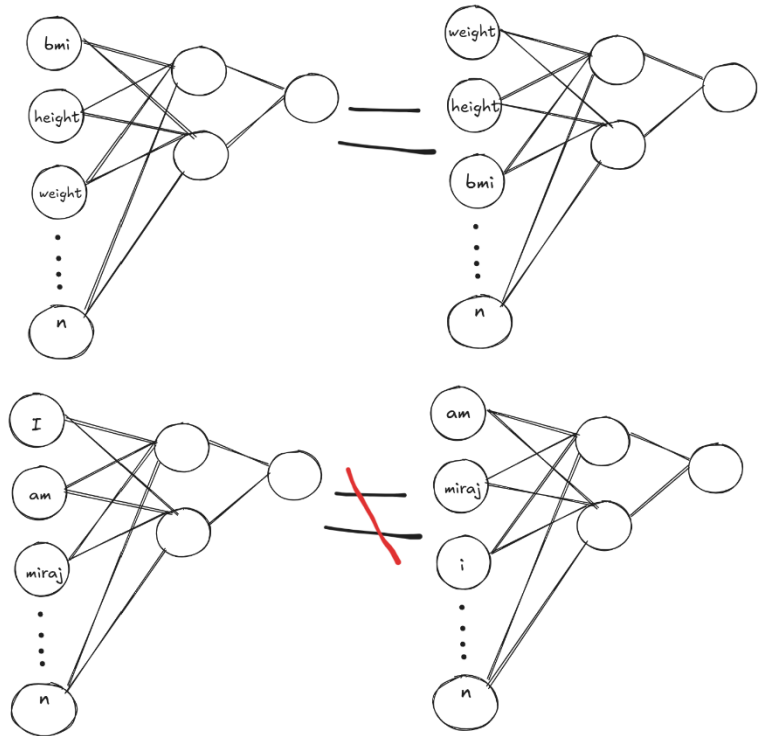
I am miraj	Neutral
Miraj is software engineer	Neutral
I love to play the guitar	Positive

Problem with ANN :

- 1) No fixed input size: Sentence lengths vary, so inputs can have different lengths. Although padding can be used to make all sequences the same size, it introduces extra padding tokens that do not carry meaningful information and may add unnecessary computation.



1) Sentence order matters in *NLP* because the meaning of a sentence depends on the sequence of its words. A standard *ANN* treats inputs as independent features and does not inherently capture these sequential relationships.



I am miraj	Neutral
Miraj is software engineer	Neutral
I love to play the guitar	Positive

Vocab = ⁰I ¹am ²miraj ³is ⁴Software ⁵engineer ⁶love ⁷to ⁸play ⁹the ¹⁰guitar

vocab_size = 11

i = [1 0 0 0 0 0 0 0 0 0 0]

am = [0 1 0 0 0 0 0 0 0 0 0]

miraj = [0 0 1 0 0 0 0 0 0 0 0]

is = [0 0 0 1 0 0 0 0 0 0 0]

software = [0 0 0 0 1 0 0 0 0 0 0]

engineer = [0 0 0 0 0 1 0 0 0 0 0]

love = [0 0 0 0 0 0 1 0 0 0 0]

to = [0 0 0 0 0 0 0 1 0 0 0]

play = [0 0 0 0 0 0 0 0 1 0 0]

guitar = [0 0 0 0 0 0 0 0 0 0 1]

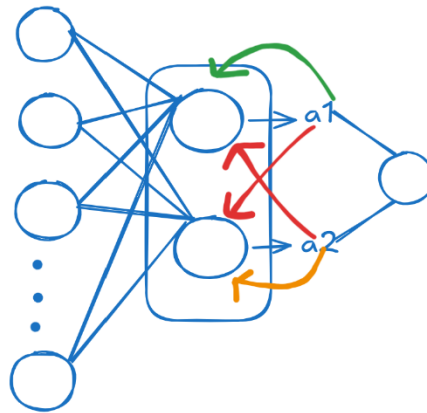
Feed Forward In RNN

[timestep, input features]

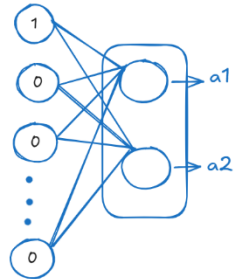
row 1 → i am miraj
[[1 0 0 0 0 0 0 0 0 0 0] [0 1 0 0 0 0 0 0 0 0 0] [0 0 1 0 0 0 0 0 0 0 0]]
t=1 t=2 t=3

(3, 11) = (no of timestep, no of features)

The Architecture



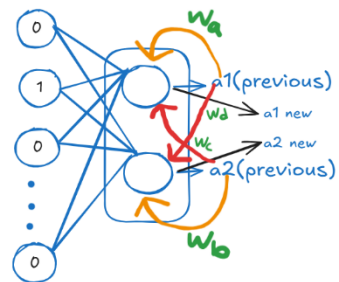
t=1



$$a1 = 1 \times w1 + 0 \times w1 + \dots + 0 \times w1 + b1$$

$$a2 = 1 \times w2 + 0 \times w2 + \dots + 0 \times w2 + b2$$

t=2



$$a1 \text{ new} = 0 \times w1 + 1 \times w1 + \dots + 0 \times w1 + b1 + a1(\text{previous}) \times w_a + a2(\text{previous}) \times w_c$$

$$a2 \text{ new} = 0 \times w2 + 1 \times w2 + \dots + 0 \times w2 + b2 + a1(\text{previous}) \times w_a + a2(\text{previous}) \times w_d$$

$t=n$

